BINARY SEARCH

```
#include <stdio.h>

int binarySearch(int arr[], int n, int x) {
   int left = 0, right = n - 1;

   while (left <= right) {
      int m = left +(right - left) / 2;
                  if(arr[m] == x) {
                        return m;
                  }
                  if(arr[m] < x) {
                        left = m + 1;
                  }
                  else {
                        right = m - 1;
                  }
   }

   return -1;
}

int main() {
   int n, x;

   scanf("%d", &n);

   int arr[n];
```

```
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    scanf("%d", &x);

    int result = binarySearch(arr, n, x);

    if (result != -1)
        printf("Element found at index %d\n", result);
    else
        printf("Element not found\n");

    return 0;
}
```